

AI/ML intern DAY 6

TASK — BODY LANDMARK DETECTION SYSTEM

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1. Introduction

This task focused on developing an AI-powered Human Body Landmark Detection System using Computer Vision and Artificial Intelligence technologies.

The project involved:

- human pose estimation
- skeletal visualization
- landmark tracking
- body posture analysis
- AI body detection

The system was implemented using:

- Python
- OpenCV
- MediaPipe

The AI model successfully detected and visualized body landmarks from human body images.

2. Objective

The objective of this module was to:

- detect human body landmarks
 - generate skeletal structures
 - analyse body posture
 - study pose estimation workflows
 - improve AI body tracking systems
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3. Technologies Used

Technology	Purpose
Python	Programming
OpenCV	Image processing
MediaPipe	Pose estimation
Artificial Intelligence	Landmark analysis
Computer Vision	Human tracking

4. AI Human Landmark Detection

The AI system detects important human body landmarks such as:

- nose
- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks are connected to generate:

- body skeletal structures
- posture visualization
- movement analysis

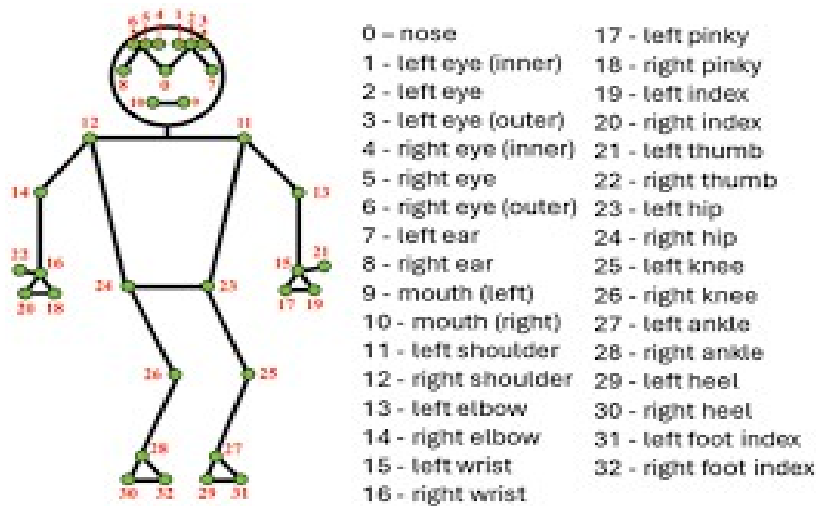


Figure 1 — AI Human Landmark Detection Output

5. Workflow of Body Detection System

The AI body detection workflow consists of several processing stages.

Stage 1 — Image Acquisition

The system receives:

- body images
 - pose photographs
 - webcam frames
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Stage 2 — Human Body Detection

The AI model detects:

- human body structure
 - body orientation
 - posture alignment
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Stage 3 — Landmark Extraction

MediaPipe extracts:

- shoulder landmarks
 - elbow landmarks
 - wrist landmarks
 - hip landmarks
 - knee landmarks
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Stage 4 — Skeletal Visualization

The system connects landmarks and generates:

- skeletal tracking
 - posture visualization
 - landmark structures
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Figure 2 — AI Skeletal Visualization

6. Landmark Detection Demo

The AI body landmark system successfully:

- detected full body posture
- tracked body landmarks
- generated skeletal connections
- visualized body structure

The system was tested using:

- standing body poses
 - arm movement positions
 - side body angles
 - multiple body orientations
-

7. Python Implementation

```
import cv2

import mediapipe as mp

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    model_complexity=2,
    min_detection_confidence=0.7
)

# Load image
image = cv2.imread(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\human.jpg"
)

# Check image
if image is None:
    print("Image not found")
    exit()

print("Image loaded successfully")

# Resize image
image = cv2.resize(image, (720, 720))

# Convert BGR to RGB
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
# Process image
results = pose.process(rgb)

# Draw landmarks
if results.pose_landmarks:

    mp_draw.draw_landmarks(
        image,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

    print("Landmarks detected successfully")

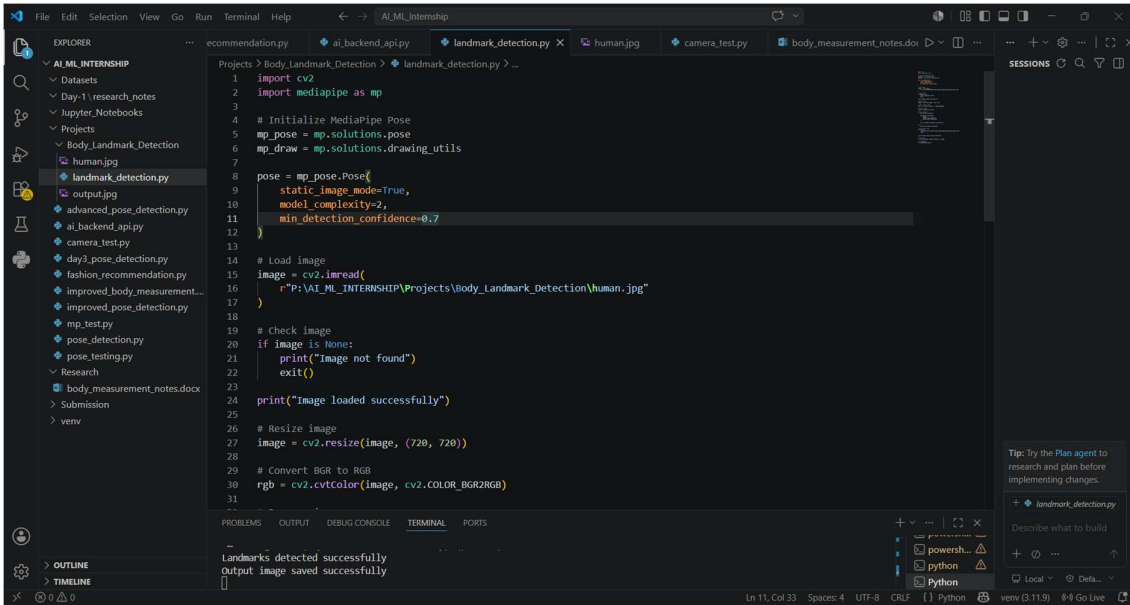
else:
    print("No landmarks detected")

# Save output image
cv2.imwrite(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\output.jpg",
    image
)

print("Output image saved successfully")

# Show image
cv2.imshow("Human Landmark Detection", image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



8. Measurement Logic

The AI system estimates body structure using landmark coordinate distances.

The system analyses:

- shoulder width
- arm length
- body posture
- body proportions

using:

- landmark coordinate calculation
- body point estimation
- skeletal alignment analysis

This measurement logic improves:

- virtual fitting systems
- AI body analysis
- posture estimation accuracy

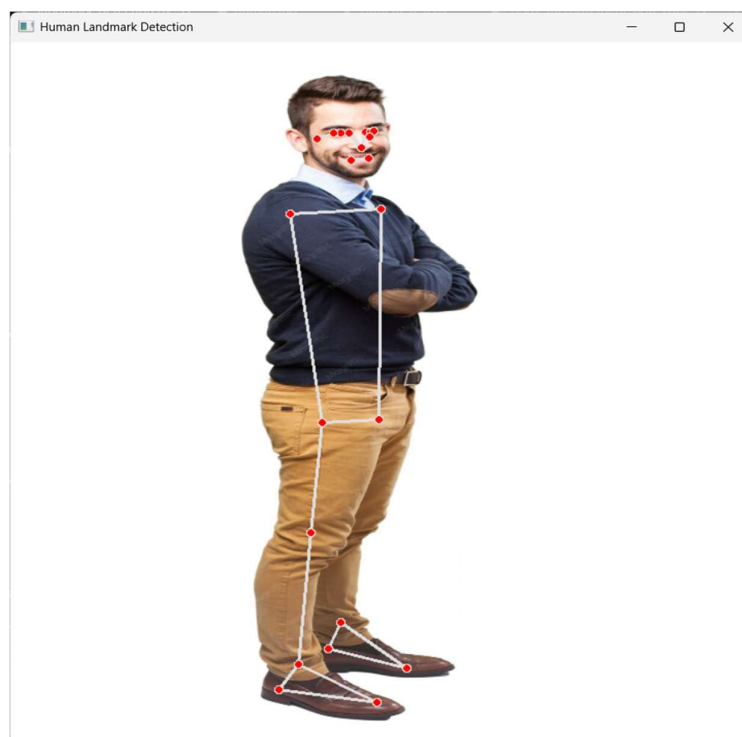
9. Testing Observations

Testing Area	Observation
Standing Pose	Accurate landmark detection
Arm Movement	Stable skeletal tracking
Side Pose	Minor landmark variation

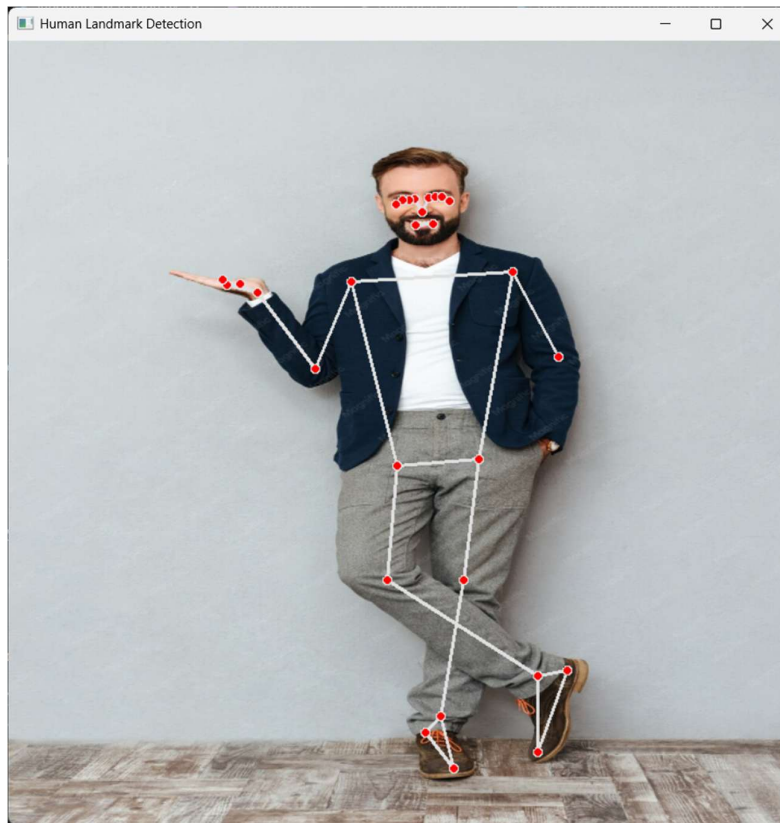
Good Lighting	High detection accuracy
Low Lighting	Reduced tracking quality



Standing Position



Side Position



Gesture

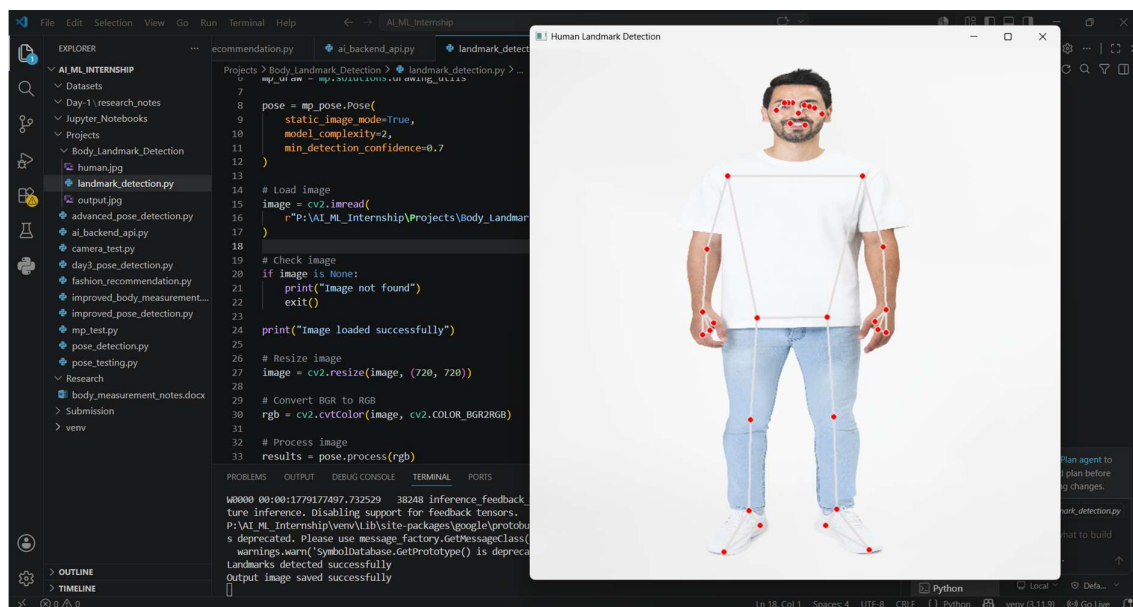


Figure 3 — AI Landmark Testing Results

10. Challenges Faced

Challenge	Impact
Poor lighting	Reduced landmark visibility
Fast body movement	Tracking instability
Complex backgrounds	Detection variation
Side body orientation	Minor pose estimation errors

11. Learning Outcomes

During this task, the following concepts were understood:

- human pose estimation
 - AI landmark tracking
 - skeletal visualization
 - body posture analysis
 - computer vision workflows
 - MediaPipe pose detection
 - AI body analysis systems
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12. Conclusion

The AI-Based Human Body Landmark Detection System successfully detected and tracked human body landmarks using Artificial Intelligence and Computer Vision technologies.

The task improved understanding of:

- AI pose estimation
- skeletal tracking systems
- landmark visualization
- body posture analysis
- AI body measurement logic
- computer vision applications

The system demonstrated how AI-based landmark detection technologies are used in:

- virtual fitting systems
- healthcare applications
- fitness tracking
- AI fashion technology
- posture analysis systems